Paper III – Subspeciality of Urology (practice paper, MCh Urology Summer 2015 set)

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III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Type: LAQ (25 marks)

Introduction

- A stent keeps the upper tract draining past obstruction, injury or an anastomosis. **Zimskind 1967** placed the first indwelling splint; **Finney 1978** introduced the self-retaining **double-J (double-pigtail)** stent.

Types and materials

Feature	Details
Size	3–8.5 Fr (adult usually 6 Fr); adults 20–30 cm by height, children about age + 10 cm
Polymers	Polyurethane, silicone (least encrusting, soft); hydrocoated silicone causes less pain
Metallic (Resonance)	Resists extrinsic compression: malignant obstruction, retroperitoneal fibrosis
Others	Hybrid metal-reinforced, thermo-expandable nitinol (Memokath), biodegradable (experimental)
Designs	Single-pigtail diversion stent; open-ended catheter to identify the ureter; modified bladder ends (Polaris loop: no benefit)

Also know: anatomy: three narrowings (PUJ (pelviureteric junction), iliac vessels, intramural ureter, the narrowest at 3–4 mm). Placement: retrograde pyelogram first, stent over a wire under fluoroscopy; antegrade route if retrograde fails.

Diagram to draw: double-J stent in situ with both coils and side holes.

Indications

- Obstruction: infected obstructed kidney (stent equals PCN (percutaneous nephrostomy)), malignancy, stricture, pregnancy.
- After URS (ureteroscopy), pyeloplasty, reimplantation, transplant; injury, leak, fistula; pre-stenting a tight ureter.

Stent symptoms and complications

- About **4 in 5** have urgency, frequency, haematuria or flank pain, measured by the **USSQ (Ureteral Stent Symptom Questionnaire)**. Cause: **intravesical material irritating the bladder and reflux up the stent**. Diameter and length matter little.
- Relief: **alpha-blockers and antimuscarinics**; intraureteral placement; shortest dwell; opioid-free ERAS (enhanced recovery after surgery) analgesia.
- **Misplacement** (submucosal) and **migration** (too short).
- **Stent failure** from compression, debris or encrustation: worst with malignant or irradiated ureter, bilateral obstruction; then PCN.
- **Encrustation** rises with dwell time, infection and stone-forming urine. **Forgotten stent** causes sepsis and renal loss; keep a register.
- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear encrustation; II circular encrustation; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.
- **Encrusted stent**: never pull by force; NCCT (non-contrast computed tomography); clear the **bladder coil**, then **ureter**, then **renal coil** (flexible URS or PCNL (percutaneous nephrolithotomy)).

Ureteroscopy

	Semirigid	Flexible
Tip	7 Fr or less	4.5–7.5 Fr; 3.6 Fr channel
Use	Distal and mid ureter	Upper ureter, kidney
Deflection	None	Over 270°

- Segura's dictum: always a safety wire. Tight ureter: stent and return after **2–4 weeks**; routine pre-stenting is unnecessary.
- **UAS (ureteral access sheath)**, usually 12/14 Fr: lowers intrarenal pressure and eases repeated passes, but **severe injury in about 1 in 8**; less after pre-stenting.
- **Ho:YAG (holmium:yttrium-aluminium-garnet, 2120 nm)** or **TFL (thulium fibre laser, about 1940 nm)**. **Dusting** (low energy, high frequency) is faster but leaves fines; **fragment and basket** confirms stone clearance and gives stone for analysis. Keep pressure and temperature low with outflow.

Steps of URS: (1) sterile urine, antibiotic dose; (2) cystoscopy and **safety guidewire**; (3) semirigid inspection, dilate only if needed; (4) access sheath over a working wire under fluoroscopy (for flexible URS); (5) low-pressure irrigation, laser, basket; (6) inspect the ureter on withdrawal; (7) decide on stent.

- **Also know:** anti-retropulsion devices; digital and single-use scopes; URS suits patients who cannot stop anticoagulants.
- Complications: sepsis about 5% (treat positive urine culture first), stricture about 3%, bleeding under 1%, perforation (stop and stent), avulsion under 1% (PCN, delayed repair).

Stent after URS?

- **Denstedt 2001** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: stent may be omitted when anatomy and renal function are normal, there is no ureteric injury and no second stage is planned. **EAU (European Association of Urology)** agrees.
- **Stent needed**: perforation, stricture or tight ureter, infection, planned second stage, solitary kidney, impacted stone.
- Removal at 3 days gave more unplanned visits than at 7 days; a string allows home removal. Stenting at SWL (shock wave lithotripsy) does not improve stone passage.

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Uncomplicated URS -> no stent (or short string stent)
Complicated       -> double-J: injury 2-6 wk; second stage until re-look
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Recent advances

- Suction sheaths, intrarenal pressure sensors, biodegradable stents.

> **Key points to remember:**
>
> - Finney 1978 double-J; symptoms from intravesical stent and reflux.
> - Encrusted stent: bladder coil, ureter, then kidney.
> - Denstedt 2001: no routine stent after uncomplicated URS.

III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR.

Type: LAQ (25 marks)

Definition and pathophysiology

- **VUR (vesicoureteral reflux)**: retrograde urine flow to the upper tract. **Primary**: deficient UVJ (ureterovesical junction) flap-valve; tunnel length to diameter should be about **5:1** (Paquin). Early budding gives a lateral orifice and dysplasia.
- **Secondary**: high bladder pressure from PUV (posterior urethral valves), neurogenic bladder or BBD (bladder and bowel dysfunction).
- In about **30% of children with UTI (urinary tract infection)** and 30% of siblings.
- **Infection, not reflux, injures the kidney.** Infected urine plus intrarenal reflux causes scars, hypertension and CKD (chronic kidney disease); some high-grade "scars" are congenital dysplasia.

International grading (VCUG, voiding cystourethrogram)

Grade	Findings
I	Non-dilated ureter only
II	Ureter, pelvis and calyces; no dilatation, sharp fornices
III	Mild–moderate dilatation of ureter and pelvis; minimal forniceal blunting
IV	Moderate dilatation and tortuosity; fornices obliterated but **papillary impressions kept**
V	Gross dilatation and tortuosity; **papillary impressions lost**; intrarenal reflux

- **Also know:** grading varies between observers; UDR (ureteral diameter ratio) predicts resolution better.

Diagram to draw: grades I–V and a Cohen cross-trigonal reimplant.

Natural history

- Resolution falls with grade and age: grade II 63–85%, grade III about 50%, IV–V rarely. **Bilateral grade IV persists most** (AUA (American Urological Association) 1997: about 10% resolve in 5 years).

Evaluation

- History of febrile UTIs and BBD; BP (blood pressure), spine, urine protein, creatinine. **RBUS (renal-bladder ultrasound)**: a normal scan does not exclude VUR.
- **VCUG**: grade and urethra. **Nuclear cystogram** (lower dose) for follow-up; **ceVUS (contrast-enhanced voiding urosonography)** avoids radiation.
- **DMSA (dimercaptosuccinic acid)**: scars and split function; a defect persisting 6 months is a scar.

Medical management

- **Aim**: prevent febrile UTI and scars while VUR resolves. **High risk**: female, infant, bilateral, grade III–V, BBD, scars, multiple febrile UTIs.
- **CAP (continuous antibiotic prophylaxis)**: one nightly dose, a quarter of the treatment dose.

Drug	Dose	Note
TMP-SMX (trimethoprim-sulfamethoxazole)	2 mg/kg TMP	Avoid under 2 months (kernicterus)
Nitrofurantoin	1–2 mg/kg	Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency; rare lung toxicity
Trimethoprim or amoxicillin	—	Young infants

- **RIVUR (2014)**, 607 children: recurrent UTI **23.6% to 12.9%** on TMP-SMX; **scarring unchanged**; most benefit with BBD.
- **Swedish Reflux Trial** (grade III–IV, age 1–2): febrile UTI in girls 19% on CAP, 23% after endoscopy, 57% on surveillance; **no difference in boys**.
- **Also know:** IRSC (International Reflux Study in Children) and Birmingham trials: surgery equalled medical care for scarring; PRIVENT: 6% absolute UTI reduction.
- **Treat BBD** (timed voiding, constipation, anticholinergics): it raises breakthrough UTI and lowers resolution and endoscopic success.
- **Circumcision**: about 4 boys with VUR circumcised prevent one UTI.
- After stopping CAP, 8–10% have a febrile UTI.
- **Guidelines**: AUA 2010 recommends CAP for VUR with BBD and treating BBD first; EAU/ESPU (European Association of Urology/European Society for Paediatric Urology) give high-risk children CAP and intervene early after breakthrough infection.
- **Intervene for** breakthrough febrile UTI, new scars, persistent high grade, non-adherence.

Case: 8-year-old boy, bilateral grade IV

- Exclude PUV, neurogenic bladder and BBD; DMSA, creatinine, BP.
- This rarely resolves at 8 years, but older boys have low UTI risk, so treating an **asymptomatic** boy is **controversial**.
- **No UTI, no scars**: observation or CAP with BBD therapy. **Breakthrough UTI, scars or falling function**: correct surgically.

Endoscopic Dx/HA (dextranomer/hyaluronic acid) injection: STING (subureteric Teflon injection, 2–3 mm below the orifice) or HIT (hydrodistension implantation, inside the orifice; better); aim for a "volcano" mound. Success per injection (Elder 2006): grade I–II about 79%, III 72%, **IV 63%**, V 51%. Day case; antibiotics 3 months, ultrasound at 2–6 weeks; 20% recurred by 2 years; late obstruction reported.

Ureteral reimplantation (open success **95.9%**, Elder 1997):

Technique	Tunnel	Note
Cohen (intravesical)	Cross-trigonal	Commonest, over 98%; one bladder opening for both sides; later ureteroscopy difficult
Politano-Leadbetter (intravesical)	New hiatus above the old one	Long tunnel; blind hiatus risks bowel injury
Glenn-Anderson (intravesical)	Advanced towards bladder neck	Short tunnel
Lich-Gregoir (extravesical)	Detrusorotomy, ureter laid in	No haematuria or spasm; **retention up to 10% after bilateral repair**
RALUR (robot-assisted laparoscopic ureteral reimplantation)	Extravesical	Less pain; results approaching open

- **Choice here: bilateral Cohen.** After surgery: ultrasound at 6–12 weeks. Complications: transient obstruction, contralateral VUR (about 1 in 10, usually resolves).

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Exclude PUV/neurogenic bladder, treat BBD
 -> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Dx/HA)
                                     No  -> observe or CAP; correct if UTI/scar
```

Key points to remember:

- Grade V loses papillary impressions; bilateral grade IV rarely resolves.
- RIVUR: CAP halves recurrent UTI but not scarring.
- Open reimplantation 95.9%; bilateral extravesical repair risks retention.

III-3a. Sacral neuromodulation

Type: SAQ (10 marks)

Definition and mechanism

- **SNM (sacral neuromodulation)** (Tanagho and Schmidt): continuous low-amplitude **S3** stimulation via a tined lead and IPG (implantable pulse generator).
- It acts on **afferents** (not motor nerves), which modulate forebrain awareness. In OAB (overactive bladder) this damps the PMC (pontine micturition centre); in retention it restores filling sensation, so the PMC switches on and the sphincter relaxes. Hence it treats **both**.

Indications (third line)

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence) 1997; urgency-frequency 1999	Bladder pain, pelvic pain
Non-obstructive retention (e.g. Fowler's syndrome) 1999; faecal incontinence 2011	Neurogenic bladder

- **Fowler's syndrome**: young women in retention with abnormal sphincter EMG (electromyography) activity; responds well to SNM.
- Retention needs some detrusor function; success falls with age.

Technique

Test phase	PNE (percutaneous nerve evaluation)	Staged tined lead
Lead	Temporary, office (dislodges)	Permanent tined, theatre
Duration	5–7 days	Up to 4 weeks
Use	OAB	More reliable; used for retention

- Prone; needle into S3 about **9 cm above the coccyx tip, 1–2 cm lateral**.

Root	Pelvic response	Foot response
S2	—	Calf and foot rotation
S3 (target)	**Bellows**, perineal sensation	**Great-toe flexion**
S4	Bellows/perineal only	None

- **Implant if at least 50% improvement** in a diary parameter (retention: 50% fewer catheterisations); IPG in the buttock.
- **Also know:** 10–16 Hz at sensory threshold; bilateral no better.

Outcomes

- **InSite**: 61% vs 42% success over medical therapy at 6 months.
- **ROSETTA**: onabotulinumtoxinA 200 U slightly better for UUI, but more UTI (urinary tract infection) (24% vs 10%) and catheterisation.

Complications

- Pain, lead migration, loss of effect, infection; **revision over 30% at 5 years**; costliest option.

Contraindications

- Failed test, inability to use the programmer, sacral anomaly.

Recent advances

- Rechargeable devices; full-body MRI (magnetic resonance imaging) conditional; PTNS (percutaneous tibial nerve stimulation) and pudendal stimulation as alternatives.

Diagram to draw: lead in S3, IPG in the buttock.

> **Key points to remember:**
>
> - Afferent modulation treats OAB and retention.
> - S3: bellows plus great-toe flexion.
> - Implant if test improves at least 50%.

III-3b. Uses of buccal mucosa in urology

Type: SAQ (10 marks)

Introduction

- **BMG (buccal mucosal graft)** is the workhorse graft for urethral reconstruction. **Humby 1941** (hypospadias); **Bürger 1992** and **El-Kasaby 1993** (stricture).

Why it works

- Thick, **non-keratinised**, hairless epithelium used to a wet environment.
- **Thin, highly vascular lamina propria**: rapid imbibition and inosculation, so take is excellent. Little contraction; low donor morbidity.
- A graft lives on its bed, so it does best on thick vascular beds.

Harvest

- Nasotracheal intubation; about 4–6 × 2–2.5 cm per cheek, **1 cm from the commissure**, **avoiding Stensen's duct** (opposite the upper second molar).
- Defat, cut about 10% oversize, fenestrate, quilt to the bed. Alternatives: **lingual**, **lip** (thinner, for the glans).
- Donor problems: pain, numbness, mouth tightness.

Uses

Technique	How	Use
Dorsal onlay (Barbagli)	Urethra rotated, dorsal urethrotomy; graft spread-fixed on the corpora	Bulbar; reliable bed
Ventral onlay	Ventral urethrotomy; graft covered by spongioplasty	Bulbar with healthy spongiosum
Dorsal inlay (Asopa)	Ventral urethrotomy, dorsal plate incised, graft inlaid	Bulbar and penile, no mobilisation
Kulkarni	One-sided dissection keeps ventral blood supply; dorsal onlay	Panurethral, one stage
Augmented anastomosis	Excise the narrowest part, anastomose the floor, graft the roof	Long dense bulbar stricture
Staged (Johanson, Bracka)	Stage 1 lay open and graft; stage 2 tubularise after 6 months or more	**Lichen sclerosus**, failed hypospadias

- **Others**: redo hypospadias, BMG perineal urethrostomy, ureteroplasty, bladder neck contracture, female urethral stricture.

Outcomes

- Bulbar BMG urethroplasty about 85–90%; dorsal equals ventral; graft equals flap; onlay beats tube.
- Results fall with time and in penile, redo and lichen sclerosus cases. **Avoid skin in lichen sclerosus** (it recurs in skin).
- **Also know:** other grafts: full-thickness skin, bladder mucosa (meatal prolapse), meshed split skin (staged only).

Recent advances

- Tissue-engineered oral mucosa.

Diagram to draw: bulbar cross-section with dorsal onlay, ventral onlay and Asopa inlay.

Key points to remember:

- Thin vascular lamina propria gives excellent take.
- Staged BMG for lichen sclerosus.
- Bulbar success about 85–90%.

III-3c. HSG in neonates

Type: SAQ (10 marks)

HSG (hysterosalpingography) is not done in neonates, so this is taken as "USG (ultrasonography) in neonates".

Introduction

- First-line, radiation-free imaging. For antenatal **UTD (urinary tract dilation)** (1–3% of pregnancies), febrile UTI (urinary tract infection), mass, poor stream.

Timing and technique

- **First scan after 48 hours**: earlier, relative dehydration under-reads dilatation.
- **Scan early if outlet obstruction is suspected**; catheterise.
- Measure **APD (anteroposterior pelvic diameter)** on a transverse mid-kidney view; assess calyces, parenchyma, ureters, bladder wall (thick if over 3 mm when full).
- Normal: **prominent hypoechoic pyramids** (not hydronephrosis); APD under 10 mm.

Grading

- **SFU (Society for Fetal Urology)**: 0 none; 1 pelvis only; 2 major calyces; 3 minor calyces too; 4 plus parenchymal thinning.
- **UTD 2014**: prenatal A1 (APD 4 to <7 mm before 28 weeks, 7 to <10 mm after) versus A2–3 (larger, or any other abnormality); postnatal:

Grade	Findings	Plan
P1	APD 10 to <15 mm or central calyces	USG at 3–6 months; no VCUG (voiding cystourethrogram) or prophylaxis
P2	APD 15 mm or more, peripheral calyces or dilated ureter	VCUG, prophylaxis optional (give if ureter >7 mm)
P3	Abnormal parenchyma or bladder	USG at 1 month, VCUG, prophylaxis, MAG3 (mercaptoacetyltriglycine) scan

Specific findings

Condition	USG
PUJO (pelviureteric junction obstruction)	Dilated pelvis, normal ureter; pyramids under 3 mm predict pyeloplasty
VUR (vesicoureteral reflux)	Normal first scan in up to 25%
PUV (posterior urethral valves)	Bilateral hydroureteronephrosis, thick bladder, dilated posterior urethra (**keyhole sign**); VCUG before discharge
Ureterocele	Thin-walled intravesical cyst
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, no central pelvis
ARPKD (autosomal recessive polycystic kidney disease)	Bilateral large, bright kidneys
Renal vein thrombosis	Large echogenic kidney, no venous flow; haematuria, mass, thrombocytopenia
Adrenal haemorrhage	Avascular, evolving suprarenal mass (neuroblastoma is vascular)

- **Also know:** mesoblastic nephroma; twinkling artefact (stones).

Recent advances

- ceVUS (contrast-enhanced voiding urosonography).

Diagram to draw: SFU grades 0–4.

> **Key points to remember:**
>
> - First scan after 48 hours, except suspected PUV.
> - UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
> - Keyhole sign suggests PUV.

III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis?

Type: SAQ (10 marks)

Introduction

- For ED (erectile dysfunction) failing less invasive treatment; satisfaction **80–90%**. Repairs: about 15% at 5 years, 30% at 10.

Types

Type	Examples	Features
Malleable	Tactra (nitinol), Genesis	Simple, cheap, no mechanical failure; always rigid
2-piece inflatable	Ambicor (discontinued)	No reservoir; poor flaccidity
3-piece inflatable	AMS 700, Titan	**Best rigidity and true flaccidity**; cylinders, scrotal pump, reservoir

- Penoscrotal or infrapubic approach; reservoir in Retzius or **ectopic submuscular** after pelvic surgery.
- **Also know:** erosion, pump migration, glans hypermobility, cylinder aneurysm; modelling in Peyronie's disease.

Biofilm

- **Definition**: a structured microbial community in a self-made **polysaccharide matrix**, attached to the implant.
- **Stages**: (1) conditioning film of host proteins coats the device within minutes; (2) reversible, then irreversible bacterial adhesion; (3) microcolonies; (4) maturation into a 3D structure with water channels and quorum sensing; (5) dispersal of free bacteria, causing clinical infection.
- **Organisms**: coagulase-negative staphylococci commonest; also S. aureus, Gram-negatives, Candida.
- **Why antibiotics fail**: the matrix blocks penetration, bacteria are slow-growing (persisters), and they survive up to **1000–1500 times** normal killing levels.
- **Relevance**: organisms from the first operation lie dormant; revising an uninfected device gives about 10% infection, cut to about 3% by **antiseptic washout**.

Prevention (infection 1–3%)

- Glucose under 200 mg/dL; treat skin and urine; decolonise MRSA (methicillin-resistant S. aureus) carriers.
- **Chlorhexidine-alcohol** skin preparation; vancomycin or cephalosporin plus gentamicin for up to 24 hours.
- **Coatings halve infection**: InhibiZone (rifampin–minocycline); hydrophilic coating dipped in antibiotic.
- **No-touch technique**: under 0.5%.

Infected implant

- Pus, fixed pump or exposed parts: remove **all** components.
- **Mulcahy salvage**: remove all parts; wash with dilute povidone-iodine (3 minutes) then saline or antibiotic; change gloves, drapes and instruments; insert a new device at once.

Success **84%**; fails with early onset, aggressive organisms, cellulitis.

Diagram to draw: 3-piece device; biofilm life cycle.

Key points to remember:

- 3-piece inflatable gives best rigidity and flaccidity.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; Mulcahy salvage 84%.

III-3e. Uretero-vaginal fistula

Type: SAQ (10 marks)

Definition and aetiology

- **UVF (uretero-vaginal fistula)**: ureter–vagina communication causing continuous leakage.
- Usually **unrecognised distal ureteric injury**; about **three-quarters follow abdominal hysterectomy**, often uneventful (ureteric injury in 0.2–2.4%). Others: obstetric, cancer, radiation.
- At risk: pelvic brim, **under the uterine artery**, beside the uterosacral ligament, vaginal fornix; by ligation, crush, thermal injury or ischaemia.

Clinical features

- **Continuous leakage with normal voiding**: the other kidney fills the bladder.
- Appears **within 4 weeks**, often with fever or flank pain, as thermal or ischaemic injury necroses.

	VVF (vesicovaginal fistula)	UVF
Voiding	Reduced	Normal
Upper tract	Normal	Hydronephrosis common
First line	Catheter, repair	Stent

Investigations

Test	Method	Reading
Double dye test	Oral phenazopyridine (orange urine); blue dye instilled into the bladder; tampon, then walk about	**Orange = UVF**; blue = VVF; both = both
Three-swab test	Blue dye in the bladder; three swabs stacked in the vagina	Blue upper swab = VVF; **wet, unstained swab = UVF**

- **CT (computed tomography) urography**, delayed phase (MR (magnetic resonance) urography if contrast is contraindicated).
- **Cystoscopy and RGP (retrograde pyelogram)**, stenting at the same sitting.
- **Exclude coexisting VVF** (up to 25% of VVF have UVF) and bilateral injury.

Management

- **Stent first**: retrograde double-J if a wire passes; otherwise PCN (percutaneous nephrostomy) and antegrade stent. **Stenting within 2 weeks** works best; remove at **4–6 weeks** and re-image.
- **Surgery** if stenting fails: **ureteroneocystostomy** with **psoas hitch** or **Boari flap** (up to 15 cm); nephropexy or ileal ureter for longer gaps; open, laparoscopic or robotic. Early repair equals delayed. Success **90% or more**.
- Follow-up ultrasound: stricture may appear up to 30 months later.

```
Leak + normal voiding -> dye test, CT urography, cystoscopy/RGP
 -> wire passes? Yes -> double-J 4-6 wk -> re-image
                 No  -> PCN + antegrade stent
 -> fails -> ureteroneocystostomy + psoas hitch / Boari
```

Prevention

- Recognise injury intraoperatively (cystoscopy); prophylactic stents only if high risk.

Diagram to draw: double dye test; psoas hitch.

Key points to remember:

- Continuous leak with normal voiding; orange tampon = UVF.
- Exclude coexisting VVF.
- Early stent; reimplant with psoas hitch if it fails.

III-3f. Prune belly syndrome

Type: SAQ (10 marks)

Definition

- **PBS (prune belly syndrome)**, Eagle–Barrett or triad syndrome: **deficient abdominal wall muscles**, **dilated poorly contracting urinary tract**, **bilateral intra-abdominal testes**.
- 1 in 29,000–44,000 births; **95% male**. **Renal dysplasia, not the abdominal wall, decides survival.**

Pathogenesis (unproven)

- Fetal **urethral obstruction**, though most have none after birth; **lateral plate mesoderm defect**; intrinsic ureteric defect; yolk sac defect. Mostly sporadic; smooth-muscle genes (MYOCD, FLNA) in families.

Features

- **Kidneys**: dysplasia in about 50%. **Ureters**: tortuous, worse distally; collagen replaces muscle, so peristalsis fails; **VUR (vesicoureteral reflux) about 75%**.
- **Bladder**: large, smooth, urachal diverticulum (urachus patent in 25–30%). **Prostatic hypoplasia**, dilated posterior urethra; atresia or megalourethra.
- **Testes**: intra-abdominal; azoospermia usual.
- **Extra-genitourinary (about 75%)**: pulmonary hypoplasia; cardiac 10%; gastrointestinal at least 30%; orthopaedic up to 65%.

Woodard classification

	Features	Outlook
I	Oligohydramnios, **pulmonary hypoplasia**, dysplasia, urethral atresia	Perinatal death
II	Full triad, moderate uropathy, **no pulmonary hypoplasia**	Risk of CKD (chronic kidney disease); debated
III	Mild, **stable function**	Good; majority

- **Also know:** RUBACE severity score (renal, ureter, bladder, abdominal wall, cryptorchidism, extra-genitourinary); "pseudo-prune" presents late.

Management

- **Neonatal**: lungs and heart first; creatinine trend, ultrasound, DMSA (dimercaptosuccinic acid) at 4–6 weeks, prophylaxis, circumcision. VCUG (voiding cystourethrogram) only if renal insufficiency or obstruction.
- Category III: surveillance. Sepsis or rising creatinine: wide **cutaneous vesicostomy**.
- **Reconstruction** after 3–6 months: ureteric tailoring and reimplantation using the healthier proximal ureter; reduction cystoplasty gives little lasting benefit.
- **Orchidopexy**: bilateral **transabdominal at about 6 months**; staged Fowler–Stephens later.
- **Abdominoplasty** (cosmesis, better cough and emptying): **Randolph** (transverse excision of the worst lower wall); **Ehrlich** (vertical incision, umbilicus kept, vest-over-pants overlap); **Monfort** (umbilicus kept, central fascial strip with lateral flaps overlapped; good exposure for combined surgery).

Prognosis

- 20% die perinatally; **up to 30% develop chronic renal failure**; nadir creatinine **under 0.7 mg/dL** is favourable.

Diagram to draw: VCUG showing large bladder and dilated posterior urethra.

> **Key points to remember:**
> - Triad; 95% male; dysplasia decides prognosis.
> - Woodard I–III.
> - Vesicostomy for drainage; orchidopexy at about 6 months.

Rapid Revision Sheet – Paper III

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)

- Finney 1978 double-J; symptoms from intravesical stent and reflux.
- Encrusted stent: bladder coil, ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS.

III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR. (LAQ)

- Grade V loses papillary impressions; bilateral grade IV rarely resolves.
- RIVUR: CAP halves recurrent UTI but not scarring.
- Open reimplantation 95.9%; bilateral extravesical repair risks retention.

III-3a. Sacral neuromodulation (SAQ)

- Afferent modulation treats OAB and retention.
- S3: bellows plus great-toe flexion.
- Implant if test improves at least 50%.

III-3b. Uses of buccal mucosa in urology (SAQ)

- Thin vascular lamina propria gives excellent take.
- Staged BMG for lichen sclerosus.
- Bulbar success about 85–90%.

III-3c. HSG in neonates (SAQ)

- First scan after 48 hours, except suspected PUV.
- UTD: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
- Keyhole sign suggests PUV.

III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis? (SAQ)

- 3-piece inflatable gives best rigidity and flaccidity.
- Biofilm: microbes in a matrix; antibiotics fail.
- Infection 1–3%; Mulcahy salvage 84%.

III-3e. Uretero-vaginal fistula (SAQ)

- Continuous leak with normal voiding; orange tampon = UVF.
- Exclude coexisting VVF.
- Early stent; reimplant with psoas hitch if it fails.

III-3f. Prune belly syndrome (SAQ)

- Triad; 95% male; dysplasia decides prognosis.
- Woodard I–III.
- Vesicostomy for drainage; orchidopexy at about 6 months.